ADAPTIVE TRAFFIC SIGNAL CONTROL LOGIC & MATHEMATICAL DERIVATION

Roadway Geometry: One-way, 2-lane roadway (180 m front school wall, 5.0 m roadway width, 5.0 m wide zebra crossing)

1. Physical Geometry & Operating Baseline

The control engine isolates and models two distinct lane functions to address student unboarding friction along the school front wall:

Parameter	Symbol	Operating Value	Units	Engineering Source / Rationale
Roadway Width	W_{road}	5.0	meters	Total 2-lane width ($2 \times 2.5\ \text{m}/lanes$).
Zebra Crosswalk Width	W_{zebra}	5.0	meters	Physical crosswalk width along the curb.
School Wall Storage	L_{wall}	180.0	meters	Physical wall length before queues spill onto arterial roads.
Left Lane Saturation	s_{left}	700.0	veh/hr/lane	Left-lane capacity constrained by student unboarding dwell friction.
Right Lane Saturation	s_{right}	1500.0	veh/hr/lane	Right-lane capacity under 20–30 km/h school-zone speed limits.
Pedestrian Capacity	s_{ped}	6000.0	peds/hr	5.0 m crosswalk capacity capped for primary student group batching.
Pedestrian Walk Speed	v_{walk}	1.2	m/s	Standard child/parent crossing velocity across the roadway.

Pedestrian Startup Time	t_{startup}	4.0	seconds	Reaction time and initial curb departure delay.
Total Lost Time / Cycle	L	8.0	seconds	Combined yellow clearance (Y_c) and all-red (AR) buffer intervals.
Operational Cycle Bounds	$C_{\min}, C_{\max}$	30.0, 120.0	seconds	Physical minimum and maximum allowable cycle duration boundaries.
Minimum Vehicle Green	$g_{v,\min}$	10.0	seconds	Enforces driver expectation and prevents rapid signal cycling.

2. Chronological Mathematical Engine

Step 1: Flow Rate Normalization (q)

Convert short-horizon 15-minute field counts (N_{15}) or rolling camera detection counts (N_W over duration W) into equivalent hourly flow rates:

$$q_{\text{left}} = N_{15,\text{left}} \times 4 \quad [\text{veh/h}]$$

$$q_{\text{right}} = N_{15,\text{right}} \times 4 \quad [\text{veh/h}]$$

$$q_p = N_{15,\text{ped}} \times 4 \quad [\text{peds/h}]$$

Step 2: Critical Lane Group Flow Ratio Evaluation (y)

Calculate lane flow ratios independently to isolate drop-off friction, then extract the critical vehicle demand ratio (y_v) using Critical Lane Group Theory (HCM 6th Edition):

$$y_{\text{left}} = \frac{q_{\text{left}}}{s_{\text{left}}}, \qquad y_{\text{right}} = \frac{q_{\text{right}}}{s_{\text{right}}}$$

$$y_v = \max\left(y_{\text{left}}, y_{\text{right}}\right)$$

$$y_p = \frac{q_p}{s_{\text{ped}}}$$

$$Y = \min\left(0.90,\ y_v + y_p\right)$$

Step 3: Pedestrian Minimum Safety Threshold ($g_{p,\min}$)

Compute the non-negotiable pedestrian walk-plus-clearance time across the 5.0 m roadway:

$$g_{p,\min} = \left\lceil t_{\text{startup}} + \frac{W_{\text{road}}}{v_{\text{walk}}} \right\rceil = \left\lceil 4.0 + \frac{5.0\ \text{m}}{1.2\ \text{m/s}} \right\rceil = \lceil 8.17 \rceil = 9\ \text{seconds}$$

Step 4: Rolling Webster Cycle Length Optimization (C)

Calculate raw Webster optimal cycle length (C_o) and clamp within physical operational bounds:

$$C_o = \frac{1.5L + 5.0}{1.0 - Y}$$

$$C_{\text{min,safe}} = L + g_{v,\min} + g_{p,\min} = 8.0 + 10.0 + 9.0 = 27.0\ \text{seconds}$$

$$C = \max\left(\max(C_{\min}, C_{\text{min,safe}}),\ \min(C_{\max}, C_o)\right)$$

Step 5: Green Split Allocation & Coupled Boundary Clamping

Deduct lost time to isolate net available green ($G_{\text{avail}} = C - L$) and split proportionally by demand ratios:

$$g'_v = G_{\text{avail}} \times \left(\frac{y_v}{Y}\right), \qquad g'_p = G_{\text{avail}} \times \left(\frac{y_p}{Y}\right)$$

Apply coupled boundary constraints to prevent timing overruns:

- **Case A (Pedestrian Underflow)**: If $g'_p < g_{p,\min}$
  $g_p = g_{p,\min}, \qquad g_v = G_{\text{avail}} - g_{p,\min}$
- **Case B (Vehicle Underflow)**: Else if $g'_v < g_{v,\min}$
  $g_v = g_{v,\min}, \qquad g_p = G_{\text{avail}} - g_{v,\min}$
- **Case C (Proportional Split)**: Else
  $g_v = \text{round}(g'_v), \qquad g_p = \text{round}(g'_p)$

Step 6: Queue Spillback Emergency Override Guard

To prevent left-lane drop-off queues from exceeding the 180 m school wall storage limit ($N_{\max} \approx 25\ \text{vehicles}$) and spilling into main arterial roads:

$$\text{If } N_{\text{queue, left}} \geq 25 \implies \begin{cases} g_v = \min\left(g_v + 15,\ C_{\max} - L - g_{p,\min}\right) \\ g_p = g_{p,\min} \end{cases}$$

3. Worked Numerical Example (Peak Window: 07:45 – 08:00 AM)

Input Data

- Left Lane Flow $q_{\text{left}} : 344 \text{ veh/h}$
- Right Lane Flow $q_{\text{right}} : 320 \text{ veh/hr}$
- Pedestrian Demand $q_p : 600 \text{ peds/hr}$

Execution Steps

1. **Flow Ratios**:
   $$y_{\text{left}} = \frac{344}{700} = 0.4914, \qquad y_{\text{right}} = \frac{320}{1500} = 0.2133$$
   $$y_v = \max(0.4914, 0.2133) = 0.4914 \quad (\text{Left Drop-Off Lane governs})$$
   $$y_p = \frac{600}{6000} = 0.1000$$
   $$Y = 0.4914 + 0.1000 = 0.5914$$
2. **Optimal Cycle (C)**:
   $$C_o = \frac{1.5(8.0) + 5.0}{1.0 - 0.5914} = \frac{17.0}{0.4086} = 41.605 \text{ seconds}$$
   $$C = \text{round}(41.605) = 42 \text{ seconds}$$
3. **Available Green (G_{avail})**:
   $$G_{\text{avail}} = 42 - 8 = 34 \text{ seconds}$$
4. **Raw Splits**:
   $$g_v' = 34 \times \left(\frac{0.4914}{0.5914}\right) = 28.25 \text{ seconds}$$
   $$g_p' = 34 \times \left(\frac{0.1000}{0.5914}\right) = 5.75 \text{ seconds}$$
5. **Safety Clamping Enforcement**:
   Since $g_p'(5.75 \text{ s}) < g_{p,\min}(9 \text{ s})$, trigger **Case A**:
   $$g_p = 9 \text{ seconds}$$
   $$g_v = 34 - 9 = 25 \text{ seconds}$$

Final Timing Output

- **Target Cycle Length (C)**: 42 s
- **Vehicle Green (g_v)**: 25 s
- **Pedestrian Green (g_p)**: 9 s
- **Total Lost Time (L)**: 8 s

4. Micro-Actuation Logic (Gap-Out & Pedestrian Interlock)

During runtime execution, calculated macro green splits (g_v, g_p) define ceiling limits ($g_{v,\max}$). Signal state transitions evaluate real-time detector headways every 0.1 seconds:

- **Rest-on-Green** : If no pedestrians are detected ($N_p = 0$), hold signal on VEHICLE_GREEN beyond g_v, bypassing g_p entirely until pedestrian demand is registered.
- **Gap-Out Early Truncation**: If $t_{\text{elapsed}} \geq g_{v,\min}$ AND measured vehicle headway gap $h_{\text{gap}} > 3.0\ \text{s}$, force immediate transition to VEHICLE_YELLOW.

Max-Out Limit: If vehicle headway remains continuous ($h_{\text{gap}} \leq 3.0\ \text{s}$), forcibly terminate vehicle green when $t_{\text{elapsed}} \geq g_v$.